COURSE CODE: CSA1712 - ARTIFICIAL INTELLIGENCE
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Question 1 – Rain and Wet Ground

Aim

To prove that the ground is wet using the **Resolution Algorithm**.

Knowledge Base

1. If it rains, then the ground becomes wet.
2. It is raining.

Goal

Prove that the ground is wet.

Step 1: Propositional Representation

Let

- R = It is raining
- W = Ground is wet

Knowledge Base:

1. $R \rightarrow W$
2. R

Goal: W

Step 2: Convert into CNF

Using $P \rightarrow Q \equiv \neg P \vee Q$

- C1 : $\neg R \vee W$
- C2 : R

Step 3: Negate the Goal

Goal = W

Negated

Goal:

- C3 : $\neg W$

Step 4: Resolution

Resolve C1 and C2:

$(\neg R \lor W), (R)$

$\Rightarrow$ C4 : W

Resolve C4 and C3:

$(W), (\neg W)$

$\Rightarrow \square$ (Empty Clause)

Conclusion

Since the **Empty Clause (□)** is obtained, the negation of the goal is false.

Therefore, the ground is wet is proved.

Question 2 – Student Assignment Submission

Aim

To prove that Rahul receives internal marks using Resolution.

Knowledge Base

1. If a student submits the assignment, then the student receives internal marks.
2. Rahul submitted the assignment.

Goal

Prove that Rahul receives internal marks.

Step 1: Propositional Representation

- S = Rahul submitted assignment
- M = Rahul receives internal marks

Knowledge Base:

1. $S \rightarrow M$
2. S

Goal: M

Step 2: CNF

- C1 : $\neg S \lor M$
- C2 : S

Step 3: Negate Goal

- C3 : $\neg M$

Step 4: Resolution

Resolve C1 and C2:

$(\neg S \lor M)$, (S)

$\Rightarrow$ C4 : M

Resolve C4 and C3:

(M), $(\neg M)$

$\Rightarrow \square$

Conclusion

Empty Clause is obtained.

Hence, Rahul receives internal marks is proved.

Question 3 – Library Membership

Aim

To prove that Priya can borrow books using Resolution.

Knowledge Base

1. If a person is a library member, then the person can borrow books.
2. Priya is a library member.

Goal

Prove that Priya can borrow books.

Step 1: Propositional Representation

- L = Priya is a library member
- B = Priya can borrow books

Knowledge Base:

1. $L \rightarrow B$
2. L

Goal: B

Step 2: CNF

- $C1 : \neg L \vee B$
- $C2 : L$

Step 3: Negate Goal

- $C3 : \neg B$

Step 4: Resolution

Resolve C1 and C2:

$(\neg L \vee B), (L)$

$\Rightarrow C4 : B$

Resolve C4 and C3:

$(B), (\neg B)$

$\Rightarrow \square$

No further clauses can be generated.

Conclusion

Empty Clause is derived.

Therefore, Priya can borrow books is proved.

Question 4 – Placement Eligibility

Aim

To prove that Arun is eligible for placement using Resolution.

Knowledge Base

1. If a student clears the aptitude test, then the student is eligible for placement.
2. Arun cleared the aptitude test.

Goal

Prove that Arun is eligible for placement.

Step 1: Propositional Representation

- A = Arun cleared aptitude test
- P = Arun is eligible for

placement Knowledge Base:

1. $A \rightarrow P$

2. A

Goal: P

Step 2: CNF

- C1 : ¬A ∨ P
- C2 : A

Step 3: Negate Goal

- C3 : ¬P

Step 4: Resolution

Resolve C1 and C2:

(¬A ∨ P), (A)

⇒ C4 : P

Resolve C4 and C3:

(P), (¬P)

⇒ □

Resolution Tree

```
¬A ∨ P
  |
  | Resolve with A
  |
  P
  |
  | Resolve with ¬P
  |
  □
```

Conclusion

Empty Clause is obtained.

Hence, Arun is eligible for placement is proved.

Question 5 – Access Control System

Aim

To prove that the user is granted access using Resolution.

Knowledge Base

1. If a user enters the correct password, then the user is authenticated.
2. If a user is authenticated, then the user is granted access.
3. The user entered the correct password.

Goal

Prove that the user is granted access.

Step 1: Propositional Representation

- P = Correct password entered
- A = User authenticated
- G = User granted access

Knowledge Base:

1. $P \rightarrow A$
2. $A \rightarrow G$
3. P

Goal: G

Step 2: CNF

- C1 : $\neg P \lor A$
- C2 : $\neg A \lor G$
- C3 : P

Step 3: Negate Goal

- C4 : $\neg G$

Step 4: Resolution

Resolution 1

$(\neg P \lor A), (P)$

$\Rightarrow$ C5 : A

Resolution 2

$(\neg A \lor G), (A)$

⇒ C6 : G

Resolution 3

(G), (¬G)

⇒ □

Conclusion

Empty Clause is obtained.

Therefore, the user is granted access is proved using the Resolution Algorithm.

Overall Result

All the given goals are successfully proved using the **Resolution Algorithm**, and in each case the derivation of the **Empty Clause (□)** confirms that the goal logically follows from the Knowledge Base. This demonstrates the use of **resolution-based automated reasoning in Artificial Intelligence**.